Contamination sweep: regime-dependent server behavior under declared attacks

Contamination sweep: regime-dependent server behavior under declared attacks I

This contamination experiment compares an active-inference colony with server operating points named for the FedGVI client-divergence vocabulary (Mildner et al. 2025); the cited sources do not make this categorical belief-fusion comparison. A colony of 7 sentinels broadcasts soft beliefs about a shared state, while 2 saboteurs mix toward a confident-wrong delta at each contamination `rate`. `KLD` is the standard log-linear pool (server robustness 0, eq. 7); the other labels denote `robust_aggregate` heuristic settings with constants KLD (c=0.00), RKL (c=1.50), AR (c=1.30), beta (c=1.70), rcce (c=1.60), not literal client losses or divergences. The estimand is consensus mass $q(\text{true state})$.

Contamination sweep: regime-dependent server behavior under declared attacks I

As the contamination rate rises across $\{0, 0.225, 0.45, 0.675, 0.9\}$, the **standard** (`KLD`) consensus accuracy degrades monotonically:

Contamination sweep: regime-dependent server behavior under declared attacks I

Table 1: Consensus accuracy $q(\text{true state})$ by contamination rate and configured server operating point ($n = 1$ deterministic sweep per cell). `KLD` is the standard log-linear pool (eq. 6); the other columns use the fixed heuristic constants listed above in the same seeded colony. As the saboteurs capture more belief mass, `KLD` falls monotonically while at least one robust member remains above the stated accuracy threshold. This table is descriptive; inferential paired evidence appears below.

Contamination rate	KLD	RKL	AR	beta	rcce
0	1.000	0.997	0.998	0.996	0.996
0.225	0.999	0.993	0.995	0.990	0.992
0.45	0.995	0.987	0.990	0.984	0.986
0.675	0.975	0.985	0.989	0.981	0.983
0.9	0.693	0.984	0.988	0.980	0.982

Contamination sweep: regime-dependent server behavior under declared attacks II

- Standard accuracy degrades monotonically with rate: Yes
- At the worst rate (0.900), at least one robust member stays at or above the accuracy threshold 0.50: Yes (standard accuracy there is 0.6928, highest pooled robust mean 0.9880; worst-rate display method beta)

The rate trend above is a single deterministic sweep per cell. To attach a per-rate paired test, each contamination rate is re-run as $n_{\mathrm{trial}} = 960$ matched trial replicates nested within the fixed seeded world, and every robust member is compared against the standard pool at that rate; the resulting p-values are BH-deflated per method ((Benjamini and Hochberg 1995); $\alpha = 0.05$). This table is a rate-resolved diagnostic, not a continuous-family proof: only the displayed method-rate cells with `Reject = Yes` survive their own per-method BH family:

Contamination sweep: regime-dependent server behavior under declared attacks I

Table 2: Per-contamination-rate standard-vs-robust paired tests (matched-pairs Wilcoxon, (Wilcoxon 1945; Fay and Proschan 2010), $n_{\mathrm{trial}} = 960$ matched trial replicates per cell), BH-deflated within each method's rate family. The trial-level result is conditional on the fixed seeded world and is not a claim about 960 independent worlds. The `KLD` baseline is excluded because it is the standard reference, not a self-contrast. The displayed d-equivalent is a rank-biserial-derived transform, not raw Cohen's d; the signed-saturation marker flags contrasts where the rank-biserial correlation saturates at ± 1. These contrasts decorate the server-side `robust_aggregate` heuristic only.

Method	Rate	Rank-biserial-derived d-equivalent	Label	Raw p	q	Reject
RKL	0	saturated (r=-1)	large	1.11e-158	1.85e-158	Yes
RKL	0.225	saturated (r=-1)	large	1.11e-158	1.85e-158	Yes
RKL	0.45	saturated (r=-1)	large	1.11e-158	1.85e-158	Yes
RKL	0.675	13.79	large	1.88e-155	2.35e-155	Yes
RKL	0.9	-0.37	small	1.06e-06	1.06e-06	Yes
AR	0	saturated (r=-1)	large	1.11e-158	1.47e-158	Yes
AR	0.225	saturated (r=-1)	large	1.11e-158	1.47e-158	Yes
AR	0.45	-303.73	large	1.13e-158	1.47e-158	Yes
AR	0.675	160.07	large	1.17e-158	1.47e-158	Yes
AR	0.9	-1.57	large	1.62e-61	1.62e-61	Yes
beta	0	saturated (r=-1)	large	1.11e-158	1.85e-158	Yes
beta	0.225	saturated (r=-1)	large	1.11e-158	1.85e-158	Yes
beta	0.45	saturated (r=-1)	large	1.11e-158	1.85e-158	Yes
beta	0.675	2.81	large	6.29e-106	7.86e-106	Yes
beta	0.9	0.61	medium	6.04e-15	6.04e-15	Yes
rcce	0	saturated (r=-1)	large	1.11e-158	1.85e-158	Yes

Contamination sweep: regime-dependent server behavior under declared attacks II

rcce	0.225	saturated (r=-1)	large	1.11e-158	1.85e-158	Yes
rcce	0.45	saturated (r=-1)	large	1.11e-158	1.85e-158	Yes
rcce	0.675	5.67	large	2.37e-141	2.96e-141	Yes
rcce	0.9	0.14	negligible	6.54e-02	6.54e-02	No

Contamination sweep: regime-dependent server behavior under declared attacks III

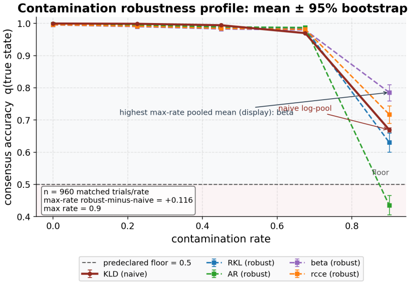

Figure 1: Consensus accuracy: probability mass assigned to the true hidden state. The plotted estimand is q(true state). Source relation: original project robustness extension; estimand: true-state probability mass; uncertainty: matched-trial percentile-bootstrap intervals over configured trials. The colony of 7 sentinel agents (acuity 0.55, of which 2 are saboteurs) as a function of contamination rate. x-axis: saboteur convex-mix contamination rate, sampled over $\{0, 0.225, 0.45, 0.675, 0.9\}$; y-axis: consensus accuracy q(true state) (probability mass on the true hidden state). One curve per configured server operating point: standard `KLD` plus the robust `robust_aggregate` settings $KLD(c = 0.00), RKL(c = 1.50), AR(c = 1.30), beta(c = 1.70), rcce(c = 1.60)$; these curves do not apply the named client losses or divergences. The dashed floor is the predeclared accuracy threshold 0.50; the in-figure box reports the matched-trial sample size and the largest-rate pooled robust-minus-naive separation, where the standard `KLD` log-linear pool reaches 0.6697 and the highest pooled robust mean reaches 0.7857. Robust means are similar to or slightly below the standard pool at low contamination and some pooled robust operating points separate in favor of the robust family under severe contamination; individual robust members can still fall below the floor at the largest rate. The linear y-axis is deliberately truncated just below the threshold band so the curves, floor, and CIs remain legible. The plotted curve is the matched-trial mean over 960 trials per rate with percentile-bootstrap 95% CIs; intervals are conditional on the fixed seeded true state and attack geometry, not alternate world models. The single-colony mechanistic table above remains descriptive and deterministic. The formal verdict-rate statistical test is reported immediately below.

Earned robustness verdict at the decisive rate I

The headline "robust beats standard" claim is *computed*, never asserted. Across 960 matched trial replicates nested within the fixed seeded world at contamination rate 0.800 — each redrawing the heterogeneous healthy colony while holding the run's true state and attack target fixed — every robust divergence's per-trial consensus accuracy is compared against the standard pool's by the matched-pairs Wilcoxon signed-rank test ((Wilcoxon 1945)), and the family of p-values is deflated with Benjamini–Hochberg FDR ((Benjamini and Hochberg 1995); $\alpha = 0.05$). A method wins if and only if BH rejects its null and its effect size is positive.

Earned robustness verdict at the decisive rate I

Table 3: Per-method paired verdict against the standard pool at verdict rate 0.800 ($n_{\mathrm{trial}} = 960$ matched trial replicates, signed-rank test under the paired-difference assumptions of sec. 5.28). This is a fixed-world conditional diagnostic; seed-level inference is reported separately in sec. 12.2.4. Effect size is the matched-pairs rank-biserial correlation; positive values mean the robust member exceeds the standard pool. A `Wins` value is true only when the BH-deflated null is rejected *and* the effect is positive, with FDR scoped to this verdict family.

Robust divergence	Effect size	Raw p	q	Wins
AR	1.0000	1.11e-158	1.11e-158	Yes
RKL	1.0000	1.11e-158	1.11e-158	Yes
beta	1.0000	1.11e-158	1.11e-158	Yes
rcce	1.0000	1.11e-158	1.11e-158	Yes

Earned robustness verdict at the decisive rate II

Table 4: Standardized-effect verdict at rate 0.800 ($n_{\mathrm{trial}} = 960$ matched trial replicates): the rank-biserial effect and its derived d-equivalent display label, the mean robust-minus-standard accuracy difference with its 95% bootstrap CI, raw and BH-deflated p-values, the observed-effect design power of the paired Wilcoxon at $\alpha = 0.05$ (alternative greater), and the prospective n_{trial} a confirmatory fixed-world replication should budget for power 0.80. The signed-saturation marker replaces a misleading literal where the rank-biserial effect saturates; the power calculation is a planning approximation, not independent evidence.

Method	Rank-biserial-derived d-equivalent	Label	Mean acc. diff	95% CI	Raw p	q	Design power	n for power	Reject
AR	saturated (r=+1)	large	0.0846	[0.0821, 0.0873]	1.11e-158	1.11e-158	1.0000	5	Yes
RKL	saturated (r=+1)	large	0.0809	[0.0783, 0.0834]	1.11e-158	1.11e-158	1.0000	5	Yes
beta	saturated (r=+1)	large	0.0764	[0.0738, 0.0790]	1.11e-158	1.11e-158	1.0000	5	Yes
rcce	saturated (r=+1)	large	0.0787	[0.0761, 0.0814]	1.11e-158	1.11e-158	1.0000	5	Yes

Earned robustness verdict at the decisive rate III

Table 5: Per-method consensus accuracy at the verdict rate 0.800 with 95% percentile-bootstrap CI, including the standard `KLD` baseline. The CI is conditional on the seeded matched-trial design and resamples trials, not alternate world models. The standard pool sits at 0.9021 ($[0.8993, 0.9049]$); the highest pooled robust mean among the configured members is 0.9867 ($[0.9865, 0.9869]$).

Method	n	Mean acc. @ verdict rate	95% CI
KLD	960	0.9021	[0.8993, 0.9049]
RKL	960	0.9829	[0.9827, 0.9832]
AR	960	0.9867	[0.9865, 0.9869]
beta	960	0.9785	[0.9782, 0.9787]
rcce	960	0.9808	[0.9806, 0.9810]

Earned robustness verdict at the decisive rate IV

- Naive pool mean accuracy at the verdict rate: 0.9021 (per-trial mean over 960 trials; the bootstrap CI is in tbl. 5)
- At least one robust method is a BH-rejected positive contrast: Yes
- Headline display method: RKL (tied set: RKL, AR, beta, rcce), rank-biserial-derived d-equivalent = saturated (r=+1) (large), mean accuracy difference 0.0846 ($[0.0821, 0.0873]$), raw $p = 1.11 \times 10^{-158}$, $q = 1.11 \times 10^{-158}$, observed-effect design power 1.0000, prospective n for power 0.80 is 5
- Headline display rule (largest positive rank-biserial effect_size; stable method order tie-break; tie-break: first robust method in divergences order): observed-effect design power 1.0000 at the run's $n_{\text{trial}} = 960$; a confirmatory replication should budget $n_{\text{trial}} = 5$ for power 0.80 (prospective $n = 7$)

Earned robustness verdict at the decisive rate I

The standard pool degrades, the robust family has conditional contrasts, and the verdict carries paired statistics with multiple-testing deflation, bootstrap intervals, and a power analysis — all produced by the statistics module, not typed into the prose.

The headline label is a deterministic display choice, not a unique scientific winner. The complete tied set is RKL, AR, beta, rcce, the largest paired mean-difference method is AR, and the worst-rate pooled display method is beta. These may differ because they answer different descriptive questions.

Server-side heuristic axis: accurate but not guaranteed I

The verdict above is reported on the **server-side heuristic axis**. fig. 2 visualizes the `robust_aggregate` divergence-reweighting that down-weights agents at pooling time. Its positive formal property is the naive-recovery limit of Theorem 5 (eq. 7, the 0 residual of sec. 7.1), and it has a scoped no-go result for a declared separable objective class, not an objective certificate. It carries *no* bounded-influence guarantee. The effect sizes, CIs, and power above decorate this heuristic's contrasts; they do not certify the per-agent generalized-Bayes guarantee, which is established separately on the per-client axis in sec. 7.6.

Server-side heuristic axis: accurate but not guaranteed I

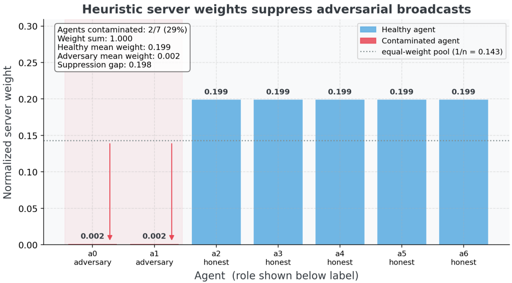

Figure 2: Server-side influence weights assigned by `robust_aggregate`. Source relation: original project server-side diagnostic; estimand: normalized pooling weight; uncertainty: deterministic single-run display. Divergence-reweighting weights for each of the 7 agents at the verdict contamination rate 0.800 (the convex-mix strength applied to each saboteur's belief — distinct from the *count* of contaminated agents, 2 of 7, reported in the in-figure box). x-axis: agent index, zero-based ($a0$ upward), with each agent's role (honest / adversary) shown beneath its label; y-axis: normalized pooling weight, with weights summing to one and the dotted reference marking the equal-weight pool ($1/n$). The 2 contaminated saboteur agents are highlighted, with downward arrows marking their suppression below the equal-weight reference. The heuristic down-weights saboteurs relative to both the equal-weight reference and the $7-2$ healthy agents. Important limitation: this is the **server-side heuristic axis only** — the reweighting is proven solely at the `robustness=0` recovery limit and does not carry the bounded-influence guarantee of the per-client FedGVI losses. Single deterministic run; no error band.

The heuristic axis above is the empirically sharp rule. The variational aggregator of sec. 5.8.2 (derived in full in sec. 11) is its objective-backed complement: each exact block update does not increase the stated free energy eq. 8, and a converged fixed point is coordinatewise stationary. Two diagnostics make the weight behavior concrete, both genuine `variational_aggregate` runs at robustness 1.50.

First, the descent. On a contaminated colony the free energy falls monotonically from 3.2458 to 2.3780 (a drop of 0.8678 over 11 block-coordinate iterations, converged: Yes), with a largest single-step *increase* of 8.88e-16 — machine zero, the numerical witness of the descent theorem (sec. 11.3).

Variational aggregator: conservative objective-backed weight control I

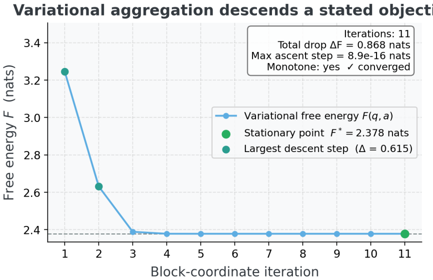

Figure 3: Variational free energy $F(q, a)$ as a function of block-coordinate descent iteration. Source relation: original project objective-descent diagnostic; estimand: free energy in nats by iteration; uncertainty: none for the deterministic seeded run. The trace is a single `variational_aggregate` fusion of a 7-agent contaminated colony (robustness 1.50). x-axis: block-coordinate iteration number; y-axis: $F(q, a)$ in nats. The curve is monotone non-increasing across all recorded iterations (largest single-step increase: 8.88×10^{-16} nats, at machine precision) and the implementation reports converged status Yes at value 2.3780 nats. This verifies objective descent on the executed run — a diagnostic the `robust_aggregate` heuristic does not provide. Deterministic seeded run; no error band.

Second, the effective-weight response. As one agent is drifted from healthy toward a confident-wrong delta, its normalized influence falls from 0.143 to below 0.001 — a factor of 267.1 below the fixed 0.143 the naive log-linear pool grants every agent regardless of how wrong it is. The gap between the falling variational curve and the flat naive line is the empirical redescending weight response, drawn.

Variational aggregator: conservative objective-backed weight control I

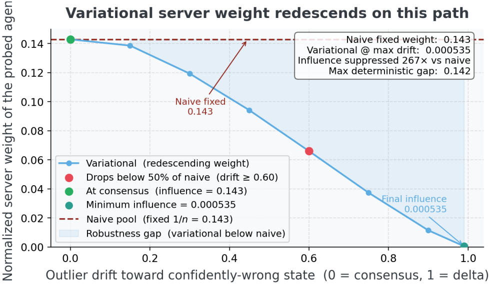

Figure 4: Normalized influence weight of one probed agent. Source relation: original project variational-server diagnostic; estimand: normalized influence weight; uncertainty: deterministic seeded sweep. The weight is shown as a function of the agent's drift toward a confident-wrong belief (delta distribution on the wrong state), under `variational_aggregate` versus the naive log-linear pool. x-axis: outlier drift — the mixing parameter carrying the probed agent's belief from the consensus posterior (zero, at consensus) to the confidently-wrong delta (one, full delta), increasing left to right; y-axis: normalized influence weight of the probed agent in the server weight vector. Under `variational_aggregate` (falling curve) the weight collapses below 0.001 as the agent goes extreme, while the naive pool holds it fixed at $1/n = 0.143$ regardless (flat line). This demonstrates redescending normalized-weight behavior on the tested path; the algebraic theorem bounds the raw effective weight, but the figure is not an estimator-level B-robustness proof. Deterministic seeded sweep over $n = 7$ agents; no error band, and no claim that the sharper `robust_aggregate` heuristic inherits this bound.

The honest trade is conservatism: because F carries the $-H(q)$ entropy term, its consensus is the maximum-entropy distribution consistent with the weighted cross-entropies, deliberately flatter than the product-of-experts. The variational aggregator therefore does *not* win the peak-accuracy verdict of sec. `sec:results-verdict` — that remains the sharp heuristic's role — and the two are reported as complements, never conflated: rigor-with-conservatism on one side, accuracy-without-an-objective on the other.